The heavy-quarkonium hyperfine splittings as the arbiter of the contact interaction

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With the ATLAS measurement of the B_c^* mass, every S-wave hyperfine splitting of the $c\bar{c}$, $b\bar{b}$ and $c\bar{b}$ ground and first-excited states is now known, together with the P-wave singlet–triplet offsets of h_c, $h_b(1P)$ and $h_b(2P)$: nine observables that test the spin–spin interaction of a heavy quark and antiquark across three reduced masses and two radial excitations. We confront them with four prescriptions for the contact term—first-order perturbation theory, a mass-independent Gaussian smearing, the mass-dependent smearing of Godfrey and Isgur, and a Yukawa form factor with the mass scale of the lattice spin–spin potential—each with a fixed or a one-loop running coupling, on top of a spin-averaged potential and quark masses fixed beforehand by the spin-independent spectrum, so that nothing but the contact prescription is tested. The data are decisive on three points. The h_c, which sits at the χ_{cJ} centroid to 0.2 MeV, limits the r.m.s. radius of any smearing to below 0.07 fm, an order of magnitude smaller than the widths used in standard potential models and small enough that smearing has no effect the data can see. The $c\bar{c}/b\bar{b}$ ratio of the $1S$ splittings, together with the $2S/1S$ ratios, fixes the short-distance strength κ of the Coulomb term, to which the spin-averaged spectrum is blind: with a running coupling the splittings require $\kappa \simeq 0.2$–0.4, consistent with $\frac{4}{3}\alpha_s$, whereas with a fixed coupling they require $\kappa = 0$. And every prescription fails in the same place: the charmonium $2S$ splitting is predicted at 55–63 MeV against the measured 48.3, the one state among the nine for which coupled-channel shifts of this size have been computed. With that state removed, first-order perturbation theory with a running coupling and no smearing describes the other eight with $\chi^2 = 2.6$ for seven degrees of freedom and a single parameter, $\Lambda = 0.21$ GeV; the smeared forms, with two or three parameters, reach $\chi^2/\mathrm{dof} = 1.1$–$2.1$. Fitted to charmonium and bottomonium alone, the prescriptions separate only in what they predict for B_c: first-order with running coupling gives $M(B_c^*) - M(B_c) = 62 \pm 2$ MeV (64.5 with the annihilation-corrected charmonium input) and $M(B_c^*(2S)) - M(B_c(2S)) = 33 \pm 1$ MeV, the smeared forms 50–58 MeV and 26–30 MeV. The ATLAS value, 64.5 ± 1.8 MeV, selects the first; the lattice values, 54(3) and 55(3) MeV, would select the second. The B_c hyperfine splitting is therefore the arbiter between contact prescriptions, and experiment and lattice QCD currently point to different ones.

1 Introduction

The spin–spin interaction between a heavy quark and antiquark is, at leading order, a contact term: one-gluon exchange gives

$$H_{SS} = \frac{32\pi\alpha_s}{9\, m_1 m_2}\, \mathbf{s}_1 \cdot \mathbf{s}_2\, \delta^3(\mathbf{r}), \tag{1}$$

so that the S-wave hyperfine splitting is proportional to $|\psi(0)|^2/m_1 m_2$ and the P-wave singlet sits at the triplet centroid. In first-order perturbation theory this works tolerably for charmonium and bottomonium separately but has always been in tension between them [1, 2], and because a δ function is singular when inserted into a Schrödinger equation, most potential models replace it by a smeared version, Gaussian with a width of order 1 GeV [2, 3], and solve the singlet and triplet non-perturbatively. The width is then a parameter, mass-independent in some models and mass-dependent in others, and its value has been set by the charmonium $1S$ splitting alone.

Until recently the data could not arbitrate. What has changed is that the B_c^* has been found [4], giving a third reduced mass; the $B_c(2S)$ doublet has been resolved [5, 6]; the $\eta_b(2S)$ exists; and the h_c and h_b masses are known to 0.1–1 MeV. Nine hyperfine observables are now measured (Table 1), spanning reduced masses from 0.9 to 2.6 GeV and two radial excitations, and they test the mass dependence, the radial dependence and the range of the contact interaction independently. Two of them deserve a caveat. The $\eta_b(2S)$ rests on a single Belle observation [7] that is omitted from the PDG summary table and was not confirmed in a later Belle search; we use it with its full, symmetrised error (± 4.2 MeV) and check that nothing depends on it. The $B_c(2S)$ splitting is not measured directly: CMS and LHCb see the $B_c^*(2S)$ only through its decay to $B_c^*\gamma$ with the photon unreconstructed, so the observed 29.1 ± 1.7 MeV separation of the two $2S$ peaks [5] is $\Delta_{1S} - \Delta_{2S}$, and $\Delta_{2S} = 35.4 \pm 2.5$ MeV follows only when combined with the ATLAS Δ_{1S}. This paper asks what they say.

The question is only sharp if nothing else is being fitted. We therefore take the spin-averaged potential $V(r)$ and the quark masses from a model-independent extraction from the spin-averaged $c\bar{c}+b\bar{b}$ spectrum [8], hold them fixed, and fit only the one to three parameters of each contact prescription to the nine splittings. Section 2 sets this up, Sec. 3 gives the comparison, Sec. 4 the consequences for B_c, and Sec. 5 concludes.

2 Set-up

Spin-averaged potential. Reference [8] represents $V(r) = -\kappa/r + S(r)$, S a cubic spline, and fits it with m_c and m_b to the eleven spin-averaged $c\bar{c}$ and $b\bar{b}$ multiplet centroids. The spectrum determines $S(r)$ to 40–70 MeV for $0.7 \le r \le 7\,\mathrm{GeV}^{-1}$ but is blind to the Coulomb coefficient: fits at $\kappa = 0$, 0.2, 0.4, 0.6 and

Table 1: The hyperfine data (MeV) [4, 5, 7]. S waves: $M(^3S_1) - M(^1S_0)$. P waves: $M(^1P_1)$ minus the $(2J+1)$-weighted centroid of the ^3P_J triplet, which vanishes for a pure contact term. Theory floors of 3 MeV (S) and 0.5 MeV (P) are added in quadrature in the fits.

observable	value	source
$c\bar{c}\,1S$	112.81 ± 0.30	$J/\psi - \eta_c$
$c\bar{c}\,2S$	48.30 ± 0.60	$\psi(2S) - \eta_c(2S)$
$b\bar{b}\,1S$	61.70 ± 2.00	$\Upsilon(1S) - \eta_b$
$b\bar{b}\,2S$	24.40 ± 4.20	$\Upsilon(2S) - \eta_b(2S)$ (Belle)
$B_c\,1S$	64.50 ± 1.80	$B_c^* - B_c$, ATLAS
$B_c\,2S$	35.40 ± 2.50	$B_c^*(2S) - B_c(2S)$, ATLAS+CMS
h_c	0.07 ± 0.15	$h_c - \bar{M}(\chi_{cJ})$
$h_b(1P)$	-0.57 ± 0.80	$h_b(1P) - \bar{M}(\chi_{bJ}(1P))$
$h_b(2P)$	-0.44 ± 1.20	$h_b(2P) - \bar{M}(\chi_{bJ}(2P))$

0.8 have $\chi^2 = 12.0$–12.2 for the same eleven centroids, with $m_c = 1.78$–1.81 and $m_b = 5.16$–5.18 GeV. These five potentials, each with its own masses, are the fixed input here, and κ is the one property of $V(r)$ that the hyperfine data, which probe $|\psi(0)|^2$, are in a position to decide.

Prescriptions. All four have the structure of Eq. (1) with $\delta^3(\mathbf{r})$ replaced by a normalised profile $\delta_X(r)$:

- *first-order*: $\delta_X = \delta^3$, the splitting evaluated in first-order perturbation theory, $\Delta = 8\alpha_s|R(0)|^2/9m_1m_2$; no P-wave shift. No free parameter beyond the coupling.
- *Gaussian*: $\delta_X = (\sigma/\sqrt{\pi})^3 e^{-\sigma^2 r^2}$, one width for all systems, as in Ref. [3]. One parameter, σ.
- *Godfrey–Isgur*: a Gaussian whose width depends on the quark masses, $\sigma_{ij}^2 = \sigma_0^2\big[\frac{1}{2} + \frac{1}{2}\big(\frac{4m_im_j}{(m_i+m_j)^2}\big)^4\big] + s^2\big(\frac{2m_im_j}{m_i+m_j}\big)^2$ [2]. Two parameters, σ_0 and s.
- *Yukawa*: $\delta_X = m^2 e^{-mr}/4\pi r$. This is the functional form with which Koma and Koma fit the non-perturbative part of the lattice spin–spin potential [9], with m the 0^{-+} glueball mass, 2.47 GeV; we use it as a one-parameter alternative to the Gaussian shape, with m free and $m = 2.47$ GeV as a check, and do not claim more lattice content for it than that.

For the three smeared forms the singlet ($\mathbf{s}_1\cdot\mathbf{s}_2 = -\frac{3}{4}$) and triplet ($+\frac{1}{4}$) radial equations are solved non-perturbatively, for S and P waves alike, with the Numerov method of Ref. [10]; the P-wave observable is then the singlet energy minus the triplet one, the spin–orbit and tensor terms cancelling in the centroid. The coupling is either one value α_s for all three systems ("fixed") or the one-loop running coupling at $\mu = \sqrt{m_1m_2}$ with Λ fitted ("running"), which makes $\alpha_s(b\bar{b})/\alpha_s(c\bar{c}) \simeq 0.68$. Each prescription is fitted at each κ; fits with the $c\bar{c}\,2S$ splitting removed, and with theory floors of 1 and 5 MeV, are the variants. Propagated errors are scaled by $\sqrt{\chi^2/\mathrm{dof}}$ where that exceeds one.

3 What the data decide

Tables 2 and 3 and Fig. 1 summarise the fits. No prescription describes all nine observables: the best χ^2 is 15–22 for six to eight degrees of freedom, and the pattern of pulls is the same for all of them. Three features of the data account for this, and each settles a question.

The h_c excludes any visible smearing. A smeared contact term shifts P-wave singlets below the triplet centroid by $\langle\delta_X\rangle_P$, which grows with the width: the h_c is predicted 16 MeV below the χ_{cJ} centroid for $\sigma = 1$ GeV and 4 MeV below for $\sigma = 2$ GeV, with α_s in each case tuned to the $c\bar{c}\,1S$ datum (Table 4, Fig. 2). The measured offset is $+0.07 \pm 0.15$ MeV; that it is consistent with zero, as a pure contact interaction requires, was noted when CLEO first measured the h_c mass precisely [11], and what follows is the quantitative version. Smearing widths of $\sigma \simeq 1$–1.2 GeV, which are what Refs. [3] and the charmonium literature use and which predict h_c masses of 3516–3517 MeV, are excluded by two orders of magnitude; the data allow a r.m.s. radius below about 0.07 fm ($\sigma \gtrsim 3.5$ GeV), the $\sigma = 3$ GeV offset of -1.6 MeV being 3σ away even with the 0.5 MeV theory floor. Every fit in Table 2 finds this on its own: the Gaussian and Yukawa widths run to $\sigma = 4.3$–4.8 GeV and $m = 7$–8 GeV, radii of 0.05–0.07 fm, where the profile is narrower than any feature of the wave functions (Fig. 3). The lattice scale $m = 2.47$ GeV is excluded for the same reason (h_c at -5 MeV). The widths do not run further for a reason worth stating: a δ function has no non-perturbative solution in three dimensions, the attractive singlet channel collapsing as the width shrinks, so the smeared prescriptions have no first-order limit. At $\sigma = 4$ GeV the non-perturbative $c\bar{c}\,1S$ splitting is already 25% above first order for the same α_s and the $b\bar{b}$ one 20% below it; at $\sigma = 8$ GeV the $c\bar{c}$ ratio is 3 and at 16 GeV it is 260. The fitted widths sit at the edge of this regime, and what the fits with smearing are really doing is using the onset of the collapse as a free parameter. Smearing, in short, is not something the data can see; it is a regulator, and the hyperfine splittings tell us to remove it.

The $c\bar{c}/b\bar{b}$ ratio fixes the Coulomb strength, given the running. The ratio of the $b\bar{b}$ to the $c\bar{c}$ $1S$ splitting, 0.547 ± 0.018, is $|R_{b\bar{b}}(0)|^2 m_c^2 \alpha_s(b\bar{b})/|R_{c\bar{c}}(0)|^2 m_b^2 \alpha_s(c\bar{c})$ in first order, and $|R(0)|^2$ for the heavier system grows faster with κ. Table 5 and Fig. 4 show the χ^2 of each prescription against the five values of κ the spectrum allows. With a fixed coupling, first-order perturbation theory wants $\kappa = 0$—no Coulomb term at all—and is excluded at any $\kappa \geq 0.2$ ($\chi^2 \geq 53$); with the running coupling, whose $\alpha_s(b\bar{b})/\alpha_s(c\bar{c}) = 0.68$ supplies two-thirds of the mass dependence, it wants $\kappa = 0.2$–0.4 and is excluded at $\kappa \geq 0.6$. That range is $\frac{4}{3}\alpha_s$ for $\alpha_s = 0.15$–0.3, the one-gluon-exchange value, which the fixed-coupling fit cannot accommodate.

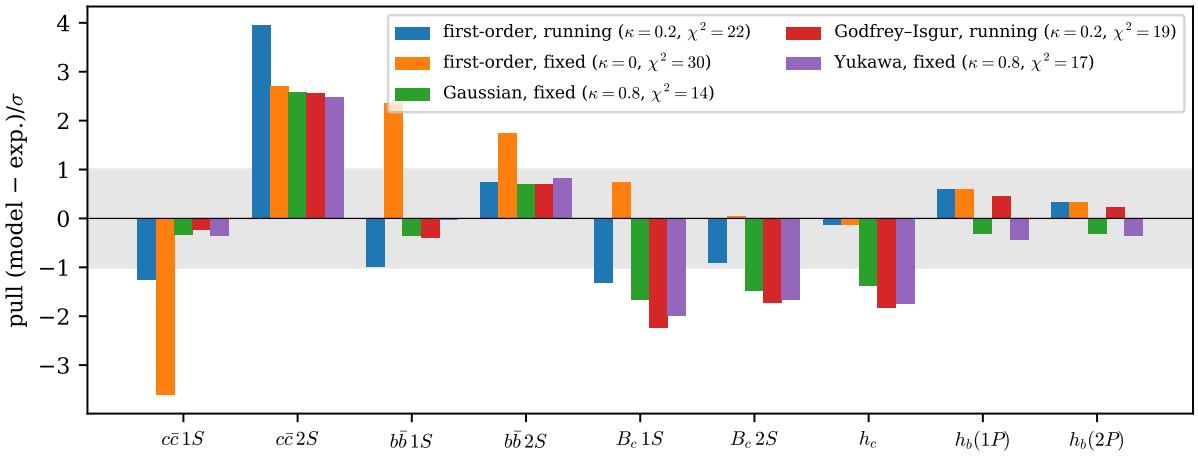


Figure 1: Pulls of the nine hyperfine observables for each prescription at the Coulomb strength κ that suits it best (Table 2). The grey band is $\pm 1\sigma$. Every prescription over-predicts the $c\bar{c}$ $2S$ splitting.

Table 2: The four prescriptions with fixed and running coupling, each at the κ of Table 5 that minimises its χ^2: fit quality for the nine observables, fitted parameters (GeV), and the r.m.s. radius of the contact term for $c\bar{c}$ (and $b\bar{b}$ where it differs). Errors scaled by $\sqrt{\chi^2/\mathrm{dof}}$.

prescription	α_s	κ	χ^2/dof	parameters	r.m.s. radius [fm]
first-order	running	0.2	21.8/8	$\Lambda = 0.19 \pm 0.01$	–
first-order	fixed	0	30.1/8	$\alpha_s = 0.33 \pm 0.01$	–
Gaussian	fixed	0.8	14.5/7	$\alpha_s = 0.28 \pm 0.03$, $\sigma = 4.38 \pm 0.80$	0.055
Gaussian	running	0	101.1/7	$\Lambda = 0.12 \pm 0.05$, $\sigma = 4.83 \pm 1.41$	0.050
Godfrey–Isgur	running	0.2	18.9/6	$\Lambda = 0.15 \pm 0.05$, $\sigma_0 = 0.20 \pm 9.14$, $s = 2.17 \pm 0.64$	0.062 / 0.022
Godfrey–Isgur	fixed	0	17.0/6	$\alpha_s = 0.27 \pm 0.04$, $\sigma_0 = 3.81 \pm 0.88$, $s = 1.44 \pm 0.77$	0.052 / 0.029
Yukawa	fixed	0.8	17.1/7	$\alpha_s = 0.27 \pm 0.03$, $m = 7.02 \pm 1.36$	0.069
Yukawa	running	0	113.9/7	$\Lambda = 0.12 \pm 0.04$, $m = 7.59 \pm 1.89$	0.064

The smeared prescriptions with a fixed coupling do the opposite, preferring $\kappa \geq 0.6$, because the non-perturbative response to the near-δ term, which enhances $c\bar{c}$ and reduces $b\bar{b}$, supplies mass dependence of the wrong sign that a stronger Coulomb term must undo; with a running coupling the mass-independent Gaussian and Yukawa forms fail at every κ ($\chi^2 \geq 80$), since both effects then act in the same direction on $b\bar{b}$, while the Godfrey–Isgur form survives by making its width mass-dependent ($s \simeq 2$), which is to say by putting the mass dependence back by hand. The short-distance strength of the potential and the running of the coupling are therefore not separately determined by the hyperfine data; they are determined together, and the physically motivated combination—$\kappa \simeq \frac{4}{3}\alpha_s$ with α_s running—is the one the data accept. The preference does not depend on the theory floor: with 5 MeV instead of 3 the first-order χ^2 runs 13.8, 8.8, 9.7, 18.9, 39.3 over $\kappa = 0$–0.8 with the running coupling and 12.5, 22.6, 39.5, 64.5, 98.2 with the fixed one (Table 5, lower block).

The $c\bar{c}$ $2S$ splitting is the universal outlier. Every fit in Table 3 predicts the $\psi(2S)$–$\eta_c(2S)$ splitting at 55–63 MeV against the measured 48.3 ± 0.6, and every $c\bar{c}$ $2S/1S$ ratio comes out 0.50–0.55 against 0.43. This is not a failure the contact prescription can fix: the ratio is set by $|R_{2S}(0)|^2/|R_{1S}(0)|^2$, which the spin-averaged potential determines, and by second-order effects that are small for any profile the h_c allows. The $b\bar{b}$ ratio, 0.40 ± 0.07, and the B_c ratio, 0.55 ± 0.04, are both reproduced (predicted 0.46–0.49 and 0.49–0.53), and the lattice $b\bar{b}$ ratio, 0.445 ± 0.028 [12], agrees with the fits better than the experimental one does; the lattice $b\bar{b}$ $1S$ splitting itself, 57.5 ± 2.5 MeV [13], is 1.3σ below experiment and 1σ from the $\kappa = 0.2$ fits. The $c\bar{c}$ $2S$ states are the only ones among the nine within 50 MeV of an open-flavour threshold, and Eichten, Lane and Quigg [14] computed the coupled-channel shifts for exactly these two: +15.7 MeV for the $\eta_c(2S)$ and −5.2 MeV for the $\psi(2S)$, reducing their splitting by 20.9 MeV. Our 12–15 MeV excess is that effect. We therefore treat the $c\bar{c}$ $2S$ splitting as a coupled-channel observable rather than a contact-term one, and refit without it.

One parameter describes the rest. With the $c\bar{c}$ $2S$ removed, first-order perturbation theory with the running coupling fits the remaining eight observables with $\chi^2 = 2.6$ for seven degrees of freedom at $\kappa = 0.2$ (8.1 at $\kappa = 0.4$), with one parameter,

Table 3: Predicted splittings (MeV) for the fits of Table 2, in the order of that table.

	$c\bar{c}\,1S$	$c\bar{c}\,2S$	$b\bar{b}\,1S$	$b\bar{b}\,2S$	$B_c\,1S$	$B_c\,2S$	h_c	$h_b(1P)$	$h_b(2P)$
experiment	112.8 ± 0.3	48.3 ± 0.6	61.7 ± 2.0	24.4 ± 4.2	64.5 ± 1.8	35.4 ± 2.5	0.1 ± 0.1	-0.6 ± 0.8	-0.4 ± 1.2
first-order, running	109.0	60.4	58.2	28.2	59.9	31.9	0.0	0.0	0.0
first-order, fixed	101.9	56.6	70.2	33.4	67.1	35.6	0.0	0.0	0.0
Gaussian, fixed	111.8	56.2	60.4	28.0	58.7	29.6	-0.7	-0.9	-0.8
Gaussian, running	115.3	58.4	36.0	15.8	47.0	23.5	-0.6	-0.6	-0.6
Godfrey–Isgur, running	112.1	56.1	60.3	28.0	56.7	28.6	-0.9	-0.1	-0.1
Godfrey–Isgur, fixed	110.3	55.9	65.8	29.6	58.2	29.0	-0.6	-0.3	-0.3
Yukawa, fixed	111.7	55.9	61.6	28.6	57.5	28.9	-0.8	-1.0	-0.9
Yukawa, running	115.4	58.0	34.5	15.0	45.6	22.6	-0.8	-0.7	-0.7

Table 4: Gaussian contact term of width σ with α_s tuned to reproduce the $c\bar{c}$ $1S$ splitting ($\kappa = 0.4$): the predicted $c\bar{c}$ $2S$ splitting and h_c offset (MeV). Measured: 48.3 and $+0.07 \pm 0.15$.

σ [GeV]	r.m.s. [fm]	α_s	$c\bar{c}\,2S$	$M(h_c) - \bar{M}(\chi_{cJ})$
0.5	0.483	2.78	33.6	−46.33
0.7	0.345	1.50	33.6	−29.29
1	0.242	0.89	39.6	−16.05
1.5	0.161	0.58	47.2	−7.28
2	0.121	0.45	51.3	−3.97
3	0.081	0.35	55.0	−1.62
4	0.060	0.30	56.4	−0.84
5	0.048	0.26	56.4	−0.49

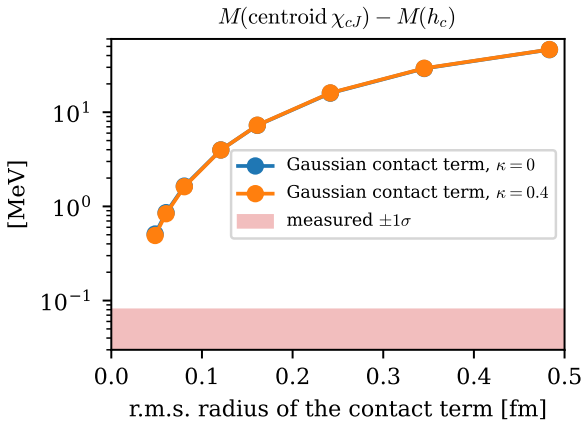


Figure 2: The h_c offset below the χ_{cJ} centroid produced by a Gaussian contact term, as a function of its r.m.s. radius, with α_s tuned to the $c\bar{c}$ $1S$ splitting; the band is the measurement.

$\Lambda = 0.213 \pm 0.011$ GeV, i.e. $\alpha_s(c\bar{c}) = 0.35$, $\alpha_s(b\bar{b}) = 0.24$, $\alpha_s(c\bar{b}) = 0.29$ (Table 6). Dropping the unconfirmed $\eta_b(2S)$ as well changes Λ by 0.001 GeV. The smeared forms do less well with more parameters: $\chi^2/\text{dof} = 6.6/6$ (Gaussian), 9.7/6 (Yukawa) and 10.7/5 (Godfrey–Isgur), the excess coming entirely from the two B_c splittings, which they under-predict by 5–7 MeV. That is the one place where the prescriptions differ in what they say, and Sec. 4 turns it into a test. One more correction is known and is not in any potential model: the η_c annihilation shift. The lattice value of the $c\bar{c}$ $1S$ splitting for the connected contribution alone is 120.3 ± 1.1 MeV [15], the difference from experiment being the $\eta_c \to gg$ shift of 7.3 ± 1.2 MeV. A contact term compared with experiment is therefore being asked to reproduce 7 MeV of annihilation physics, and the right comparison for it is the connected value. Fitting to 120.1 ± 1.2 instead of 112.8 gives $\chi^2 = 2.1/7$ with $\Lambda = 0.236 \pm 0.011$ GeV (last row of Table 6); the $b\bar{b}$ splittings are unchanged and the B_c ones move up by 2–3 MeV.

4 The B_c hyperfine splittings

The B_c is the system where the prescriptions differ most, because it is the only one whose reduced mass is not used to set their parameters, and it is where experiment and lattice QCD disagree: ATLAS finds $M(B_c^*) - M(B_c) = 64.5 \pm 1.8$ MeV [4], HPQCD predicted 54 ± 3 [16] and Mathur, Padmanath and Mondal 55 ± 3 [17], a 3σ difference. To make a prediction rather than a fit, block (a) of Table 6 repeats each acceptable fit with the two B_c splittings removed from the data, so that its parameters are set by charmonium and bottomonium alone and the $c\bar{b}$ system is interpolated. Every prescription then describes the six remaining equal-mass observables ($\chi^2/\text{dof} \leq 0.5$); they are indistinguishable on $c\bar{c}$ and $b\bar{b}$ and differ only in B_c. First-order perturbation theory with the running coupling predicts $\Delta_{1S} = 61.6 \pm 1.3$ at $\kappa = 0.2$ and 62.8 ± 1.6 MeV at $\kappa = 0.4$, and $\Delta_{2S} = 32.8$–33.6 MeV. The smeared forms predict less, and with larger errors: 57.6 ± 2.7 (Gaussian), 52.5 ± 9.4 (Godfrey–Isgur) and 50.0 ± 4.8 MeV (Yukawa) for Δ_{1S}, because the nonperturbative enhancement of the singlet, which grows with the compactness of the wave function, is tuned away by the fit for $c\bar{c}$ and $b\bar{b}$ separately but not for the intermediate $c\bar{b}$ system. Including the B_c data in the fit (block (b)) moves the first-order prediction by 0.5 MeV and pulls the smeared ones up by 2–8 MeV at the cost of their χ^2. Table 7 gives the remaining systematics of the first-order prediction: κ within the range the data allow (+1.2 MeV), the quark masses (±0.6), and the η_c annihilation correction to the char-

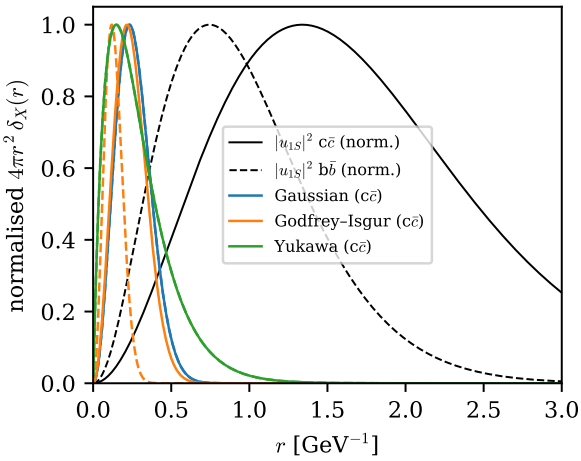


Figure 3: The fitted contact-term profiles $4\pi r^2\delta_X(r)$ (fixed coupling, $\kappa = 0$) compared with the $1S$ wave functions. The fitted widths are far inside the wave functions.

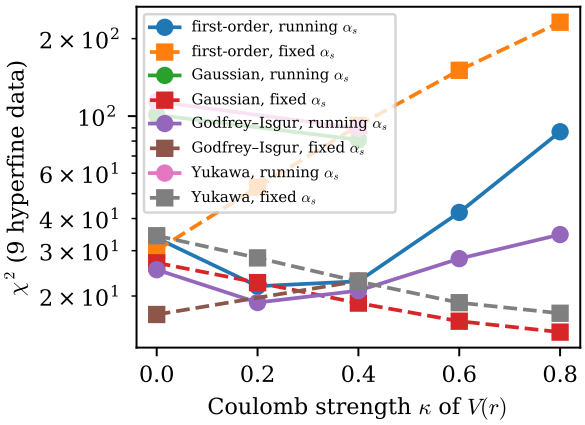


Figure 4: The χ^2 of Table 5 against κ. Circles and solid lines: running coupling; squares and dashed lines: fixed.

Table 5: χ^2 of each prescription (nine observables) against the Coulomb strength κ of the spin-averaged potential; lower block, the first-order fits with a 5 MeV theory floor. The spin-averaged spectrum has $\chi^2 = 12.0$–12.2 at every κ.

prescription	α_s	$\kappa = 0$	0.2	0.4	0.6	0.8
first-order	running	33.6	21.8	22.8	42.2	86.9
first-order	fixed	30.1	53.0	92.0	150.7	231.6
Gaussian	fixed	26.9	22.5	18.8	16.0	14.5
Gaussian	running	101.1	–	80.9	–	–
Godfrey–Isgur	running	25.3	18.9	21.0	27.9	34.7
Godfrey–Isgur	fixed	17.0	–	22.9	–	–
Yukawa	fixed	34.3	28.1	22.8	18.8	17.1
Yukawa	running	113.9	–	89.6	–	–
first-order, floor 5 MeV	running	13.8	8.8	9.7	18.9	39.3
first-order, floor 5 MeV	fixed	12.5	22.6	39.5	64.5	98.2

monium input (+2.9). Taking $\kappa = 0.2$ with the measured inputs as central and adding the κ and mass variations in quadrature,

$$M(B_c^*) - M(B_c) = 62 \pm 1 \pm 1\,\text{MeV}, \quad (2)$$

$$M(B_c^*(2S)) - M(B_c(2S)) = 33 \pm 1 \pm 1\,\text{MeV}, \quad (3)$$

$$\Delta_{1S} - \Delta_{2S} = 29 \pm 1\,\text{MeV}, \quad (4)$$

or 64.5, 34.4 and 30 MeV with the annihilation-corrected charmonium input, against the measured 64.5 ± 1.8, 35.4 ± 2.5 and 29.1 ± 1.7. The first-order prediction agrees with ATLAS at 1.3σ with the measured charmonium input and at 0.0σ with the annihilation-corrected one, and lies 2–3σ above the lattice values; the smeared predictions agree with the lattice and lie 2–3σ below ATLAS. The B_c hyperfine splitting thus discriminates between contact prescriptions that the equal-mass systems cannot separate, and the measurement and the lattice at present select different ones. If ATLAS stands, the contact interaction is the unsmeared one-gluon-exchange term with a running coupling; if the lattice value stands, the hyperfine data would be telling us that the non-perturbative singlet enhancement of a near-δ term is real physics rather than an artefact of the regulator, which would be surprising given Sec. 3. A confirmation of the ATLAS B_c^* by CMS or LHCb, or a lattice determination with the B_c at the few-MeV level of the current $b\bar{b}$ result [13], would settle it. The B_c P-wave singlet–triplet offset is predicted to vanish to 0.5 MeV by every prescription.

5 Conclusions

The nine measured heavy-quarkonium hyperfine observables, confronted with four contact-interaction prescriptions on top of a spin-averaged potential fixed by the spectrum, decide the following:

1. The h_c offset from the χ_{cJ} centroid, 0.07 ± 0.15 MeV, limits the r.m.s. radius of any smeared contact term to below 0.07 fm. The $\sigma \simeq 1$ GeV Gaussians of the potential-model literature are excluded by two orders of magnitude, and the fits themselves push every smearing to the point where it is invisible.

2. With a one-loop running coupling, the $c\bar{c}/b\bar{b}$ hyperfine ratio selects a Coulomb strength $\kappa = 0.2$–0.4 in the spin-averaged potential, consistent with $\frac{4}{3}\alpha_s$ and invisible to the spectrum itself; with a fixed coupling it selects $\kappa = 0$. Running and Coulomb strength are determined jointly, and only their one-gluon-exchange combination fits.

3. The $c\bar{c}$ $2S$ splitting is over-predicted by 12–15 MeV by every prescription, the size of the coupled-channel shift computed for the $\eta_c(2S)$ and $\psi(2S)$; it should not be used to tune contact interactions.

4. With it removed, first-order perturbation theory with a running coupling and no smearing describes the other eight observables with $\chi^2 = 2.6/7$ and one parameter; the smeared forms reach

Table 6: B_c hyperfine splittings (MeV). Upper block (a): fits to the charmonium and bottomonium data only ($c\bar{c}$ $2S$ excluded), so that the B_c columns are predictions; lower block (b): the same fits including the two B_c splittings. Errors are the propagated fit errors scaled by $\sqrt{\chi^2/\mathrm{dof}}$. Experiment: $\Delta_{1S} = 64.5 \pm 1.8$ [4], $\Delta_{2S} = 35.4 \pm 2.5$ (ATLAS+CMS), $\Delta_{1S} - \Delta_{2S} = 29.1 \pm 1.7$ (CMS [5]), 30.9 (LHCb [6]); lattice $\Delta_{1S} = 54 \pm 3$ [16], 55 ± 3 [17]. The last two columns are the predicted $c\bar{c}$ $2S$ splitting and B_c P-wave offset.

fit	χ^2/dof	Δ_{1S}	Δ_{2S}	$\Delta_{1S} - \Delta_{2S}$	$M(B_c^*)$	$c\bar{c}\,2S$	$B_c\,1P$
first-order, running, $\kappa = 0.2$	1.6/5	61.6 ± 1.3	32.8 ± 0.7	28.8 ± 0.6	6336.1	62.7	+0.0
first-order, running, $\kappa = 0.4$	7.7/5	62.8 ± 1.6	33.6 ± 0.9	29.2 ± 0.8	6337.3	60.9	+0.0
first-order, running, $\kappa = 0.2$, lattice $c\bar{c}\,1S$	1.7/5	64.5 ± 1.3	34.4 ± 0.7	30.2 ± 0.6	6339.0	66.3	+0.0
Gaussian, fixed, $\kappa = 0.8$	2.0/4	57.6 ± 2.7	29.5 ± 0.9	28.1 ± 1.9	6332.1	57.0	-0.3
Yukawa, fixed, $\kappa = 0.8$	1.5/4	50.0 ± 4.8	26.1 ± 2.3	23.8 ± 2.6	6324.5	58.0	-0.1
Godfrey–Isgur, running, $\kappa = 0.2$	1.6/3	52.5 ± 9.4	27.1 ± 4.3	25.4 ± 5.1	6327.0	58.3	-0.1
first-order, running, $\kappa = 0.2$	2.6/7	62.1 ± 1.2	33.1 ± 0.6	29.0 ± 0.5	6336.6	63.2	+0.0
first-order, running, $\kappa = 0.4$	8.1/7	63.1 ± 1.3	33.8 ± 0.7	29.3 ± 0.6	6337.6	61.3	+0.0
first-order, running, $\kappa = 0.2$, lattice $c\bar{c}\,1S$	1.8/7	64.6 ± 1.2	34.4 ± 0.6	30.2 ± 0.6	6339.1	66.4	+0.0
Gaussian, fixed, $\kappa = 0.8$	6.6/6	59.5 ± 1.7	30.2 ± 0.6	29.4 ± 1.3	6334.0	57.6	-0.5
Yukawa, fixed, $\kappa = 0.8$	9.7/6	58.1 ± 2.3	29.3 ± 0.7	28.8 ± 1.7	6332.6	57.4	-0.6
Godfrey–Isgur, running, $\kappa = 0.2$	10.7/5	57.1 ± 3.1	29.0 ± 1.5	28.1 ± 1.8	6331.6	58.0	-0.2

Table 7: Shifts (MeV) of the first-order, running-coupling prediction for the B_c splittings (block (a) of Table 6, $\kappa = 0.2$) under the systematic variations considered. The quark-mass shifts are the 1σ errors of the centroid extraction with $V(r)$ held fixed.

variation	$\delta\Delta_{1S}$	$\delta\Delta_{2S}$
$\kappa = 0.4$ instead of 0.2	+1.2	+0.8
$m_c + 0.024\,\mathrm{GeV}$	+0.6	+0.3
$m_c - 0.024\,\mathrm{GeV}$	−0.6	−0.3
$m_b + 0.023\,\mathrm{GeV}$	−0.2	−0.1
$m_b - 0.023\,\mathrm{GeV}$	+0.2	+0.1
lattice (connected) $c\bar{c}\,1S$ input	+2.9	+1.5

χ^2/dof = 1.1–2.1 with two or three, their excess coming from the B_c.

5. Fitted to charmonium and bottomonium alone, the prescriptions predict different B_c splittings: $62 \pm 2\,\mathrm{MeV}$ (64.5 with the η_c annihilation shift taken out of the charmonium input) for the unsmeared running-coupling term, 50–58 MeV for the smeared ones. ATLAS's 64.5 ± 1.8 selects the former, the lattice's 54–55 ± 3 would select the latter. The B_c hyperfine splitting is the arbiter, and experiment and lattice QCD currently disagree about the verdict.

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Code and data availability

The code, all fits and the scripts that regenerate every table and figure are available at `https://github.com/apoppong24/bc-hyperfine`; the spin-averaged potentials are those of Ref. [8].

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